

Nighttime Lights Compositing Using the VIIRS Day-Night Band: Preliminary Results

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Abstract:

The Suomi National Polar-orbiting Partnership (SNPP) Visible Infrared Imaging Radiometer Suite (VIIRS) represents a major advance in low-light imaging over the previous data sources. Building on 18 years of experience compositing nighttime data from the Defense Meteorological Satellite Program (DMSP) Operational Linescan System (OLS), NOAA's NGDC Earth Observation Group has started adapting their algorithms to process these new data. The concept of compositing nighttime data comprises combining only high quality data components over a period of time to improve sensitivity and coverage. For this work, flag image are compiled to describe image quality. The flag categories include: daytime, twilight, stray light, lunar illuminance, noisy edge of scan data, clouds, and no data. High quality data is defined as not having any of these attributes present. Two methods of reprojection are necessary due to data collection characteristics. Custom algorithms have been created to terrain-correct and reproject all data to a common 15 arc second grid. Results of compositing over two time periods in 2012 are presented to demonstrate data quality and initial capabilities. These data can be downloaded at http://www.ngdc.noaa.gov/eog/viirs/download_viirs_ntl.html.

Keywords: VIIRS DNB, Nighttime Lights

1. Introduction

On October 28, 2011 the Suomi NPP satellite launched carrying the VIIRS sensor. The low-light imaging day/night band (DNB) draws its heritage from the OLS visible band sensors, which have been flown on the DMSP platforms since the 1970's. Essentially unchanged in design since the 1970's, the OLS visible band collects global data at a spatial resolution of 2.7 km, has a 6-bit dynamic range, and is an uncalibrated imager by design. The DNB is a huge leap forward in capability from the OLS, with global data at 742m spatial resolution, a 14-bit dynamic range, and is a calibrated radiometer. Both the DNB and OLS visible bands have a comparable broad spectral range of 0.5-0.9 μ m centered at 0.7 μ m and they have the ability to collect low-light imagery at night.

Since 1996, the Earth Observation Group at NOAA/NGDC has been generating global annual nighttime lights composites using the DMSP-OLS data [1,2]. While these products have proved valuable to the scientific community, enabling the study of lighting patterns over time, and allowing researchers to study socio-economic parameters for which nighttime lights serve as a reasonable proxy, the limitations of the OLS have hindered more widespread use of these datasets. The two most troublesome of the OLS limitations are the 6-bit dynamic range, which results in the saturation of urban centers, and the lack of onboard calibration. When the VIIRS DNB data became available in early 2012, the authors started modifying existing and creating new algorithms to process these new data.

This paper details the methodology used to create the first global nighttime lights composites using the VIIRS DNB data and shows preliminary results from those initial composites.

2. Methods

Compositing algorithms for the NPP VIIRS DNB are adapted from heritage DMSP OLS algorithms, developed at NOAA NGDC [1,2], with adjustments for sensor differences and data specific parameters.

2.1. Flag Images

To enable the selection of only high-quality cloud-free nighttime data for inclusion in the composite product, pre-processing is done on the input VIIRS DNB and M15 bands to create flag images. Made as companions to the DNB aggregates, flag images are used to place the DNB pixels into classes. Flag images are 32-bit and are processed bit-wise, so each pixel can belong to more than one class by turning specific bits on. The flag

categories used in the composite processing are: daytime, terminator zone (twilight), stray light, zero lunar illuminance, no data (includes noisy edge data) and clouds present.

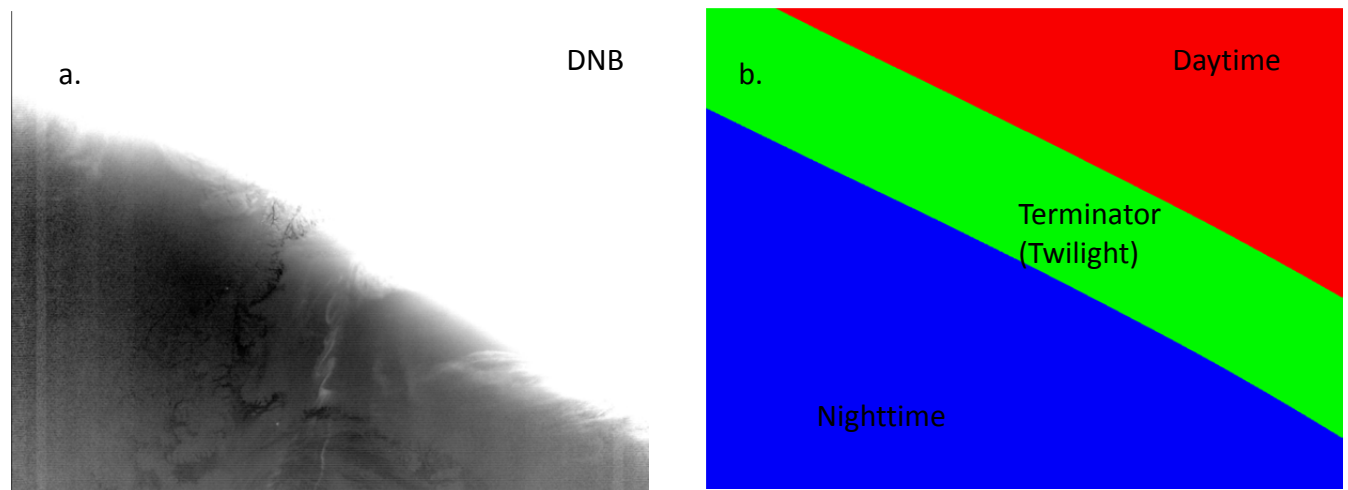


Figure 1. (a) VIIRS DNB aggregate covering the transition from nighttime to daytime.
(b) Corresponding DNB Flag image showing the daytime region in red, the terminator region in green, and the nighttime data in blue.

The daytime and terminator flags are set based on solar zenith angles. Solar zenith angles are provided as a layer in the VIIRS DNB geolocation file. The daytime flag bit is set for solar zenith angles less than 96. The terminator flag bit is set for solar zenith angles between 96 and 101. This region of the nighttime DNB imagery covers the terminator, or the transition zone from nighttime to daytime, and is of reduced quality as compared to the darker nighttime data (Figure 1).

The VIIRS DNB sensor is affected by stray light. Stray light is unintended light entering the optical path resulting in an overall increase in the recorded radiance values. A stray light flag is set using the ground-based solar zenith angles as a proxy for spacecraft solar zenith angles. In this case, an entire scan line is flagged as being subject to stray light contamination if the solar zenith angle at nadir is between 90 and 118.5.

To keep the ground pixel footprint at a nearly constant 742m, the DNB sensor has “sub-pixel” detectors that are blended, or aggregated, to make the DNB “pixel”. The aggregation scheme reduces the number of sub-pixel detectors as it moves away from nadir, resulting in 32 different aggregation zones on each side of nadir (Table 1)

Table 1. VIIRS DNB aggregation zones from nadir to edge of scan [3].

Aggregation Mode from Nadir	Number of Sub-pixels per Pixel		Number of Pixels per Mode	Time per Pixel (sec)
	Track Direction	Scan Direction		
1	42	66	184	2.5326E-04
2	42	64	72	2.4559E-04
3	41	62	88	2.3791E-04
4	40	59	72	2.2640E-04
5	39	55	80	2.1105E-04
6	38	52	72	1.9954E-04
7	37	49	64	1.8803E-04
8	36	46	64	1.7652E-04
9	35	43	64	1.6500E-04
10	34	40	64	1.5349E-04
11	33	38	64	1.4582E-04
12	32	35	80	1.3431E-04
13	31	33	56	1.2663E-04
14	30	30	80	1.1512E-04
15	29	28	72	1.0744E-04
16	28	26	72	9.9770E-05
17	27	24	72	9.2095E-05
18	27	23	32	8.8258E-05
19	26	22	48	8.4421E-05
20	26	21	32	8.0583E-05
21	25	20	48	7.6746E-05
22	25	19	40	7.2909E-05
23	24	18	56	6.9071E-05
24	24	17	40	6.5234E-05
25	23	16	72	6.1397E-05
26	23	15	24	5.7559E-05
27	22	15	32	5.7559E-05
28	22	14	64	5.3722E-05
29	21	13	64	4.9885E-05
30	21	12	64	4.6048E-05
31	20	12	16	4.6048E-05
32	20	11	80	4.2210E-05
Total			2032	

The last 4 aggregation zones at the edges of each scan show a visible increase in noise and are flagged as “no data” so they won’t be included in the DNB composite image (Figure 2).

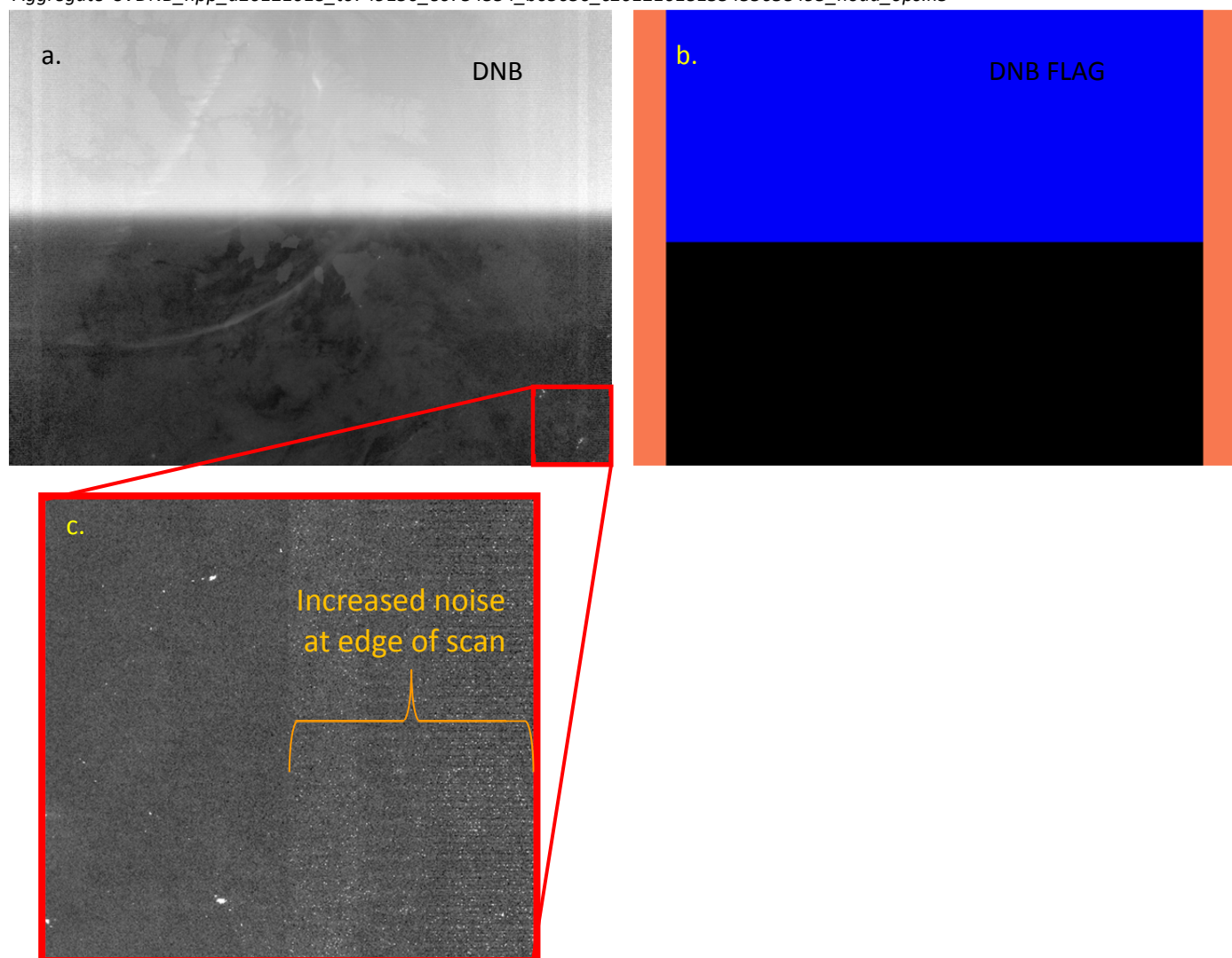


Figure 2. (a) VIIRS DNB aggregate showing the effects of stray light in the bright upper part of the image. **(b)** Corresponding DNB Flag image showing the stray light region in blue, and the noisy outer aggregation zones 29-32 in orange. **(c)** Detailed view of the apparent noise in aggregation zones 29-32.

To set the zero lunar illuminance flag, lunar illuminance values are computed using an algorithm obtained from the US Naval Observatory [4]. This algorithm approximates the lunar illuminance present at the earth's surface as a function of lunar phase, azimuth, and elevation. The lunar phase, azimuth, and elevation are in turn computed from the latitude and longitude of each DNB pixel and the time at the nadir pixel of each scan. The zero lunar illuminance flag is set when the returned lunar illuminance is less than 0.0005 lux.

At times there are fill scans within a DNB aggregate. These scans contain no real data, but are present as placeholders in the files. The “no data” flag is set in the flag image to correspond to these regions.

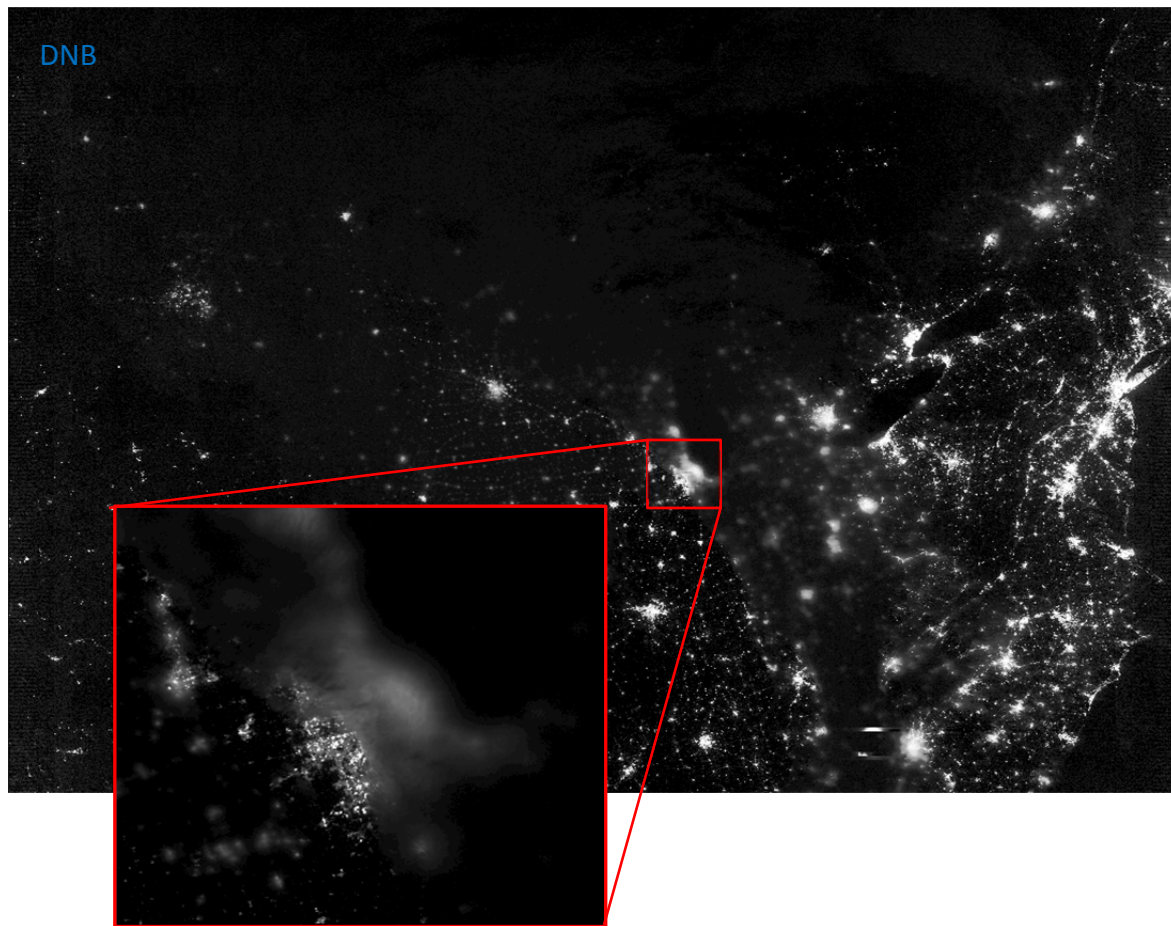
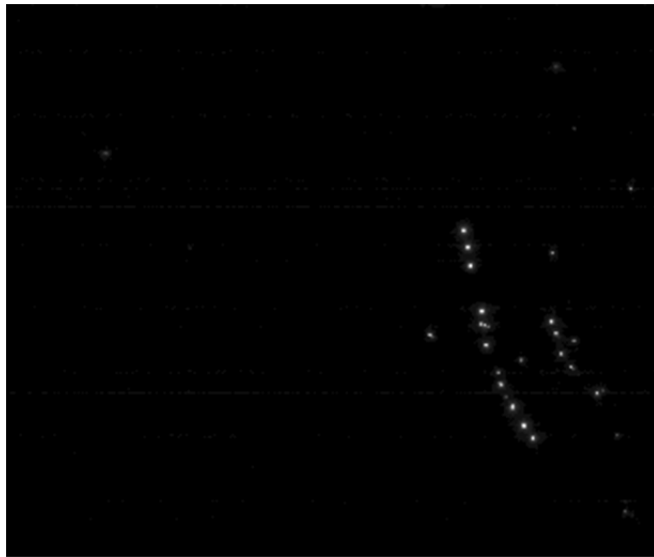
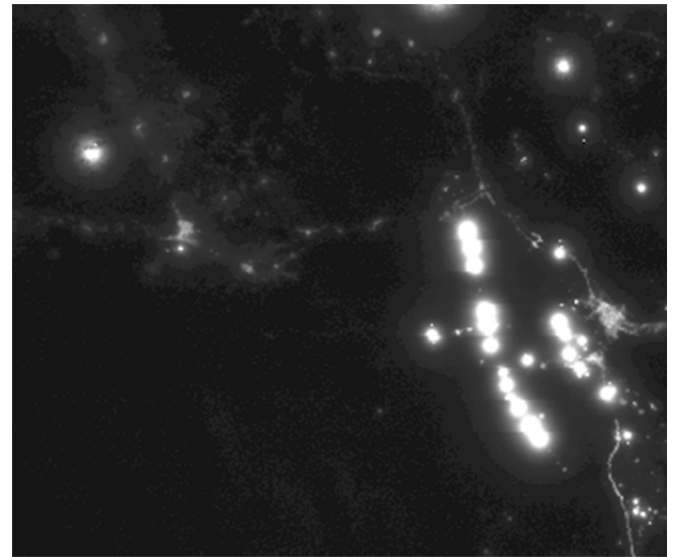


Figure 3. VIIRS DNB aggregate over the United States. Clouds are impacting nighttime lights of Chicago by reducing the intensity of the lights and by blurring the spatial detail of the light features.

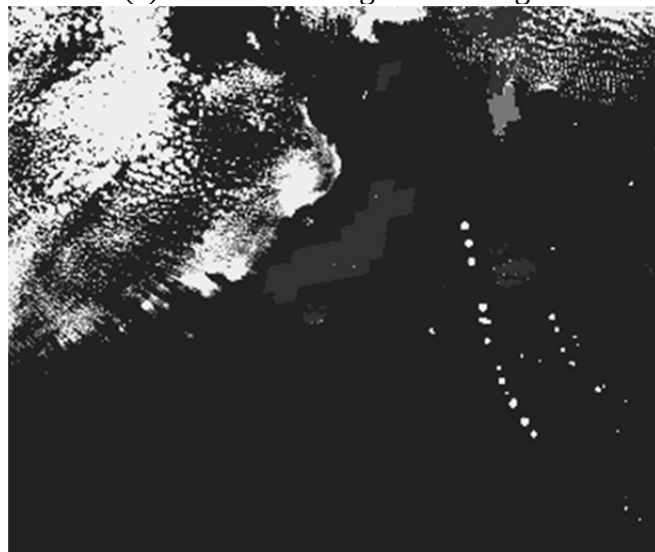
The final flag used in this prototype version is the cloud flag. It is desirable to include only cloud-free data as the presence of clouds affects both the intensity and location of lights in DNB imagery (Figure 3). Thick clouds can obscure a light completely, while thinner clouds diffuse the signal, making lights appear larger but dimmer than they would have on a clear night. To determine co-incident cloud-cover to the VIIRS DNB, the authors investigated the use of the Joint Polar Satellite Systems (JPSS) retained intermediate product cloud mask, abbreviated “IICMO”. The IICMO, as a retained intermediate product, was chosen for evaluation over the VIIRS Cloud Cover Layer Environmental Data Record (EDR) as it retains the original spatial resolution of the input VIIRS data. The IICMO cloud mask layer, however, tends to recognize gas flares as clouds, so it was not selected for making this prototype nighttime light composite (Figure 4). The authors instead converted the algorithm developed for DMSP OLS [1].



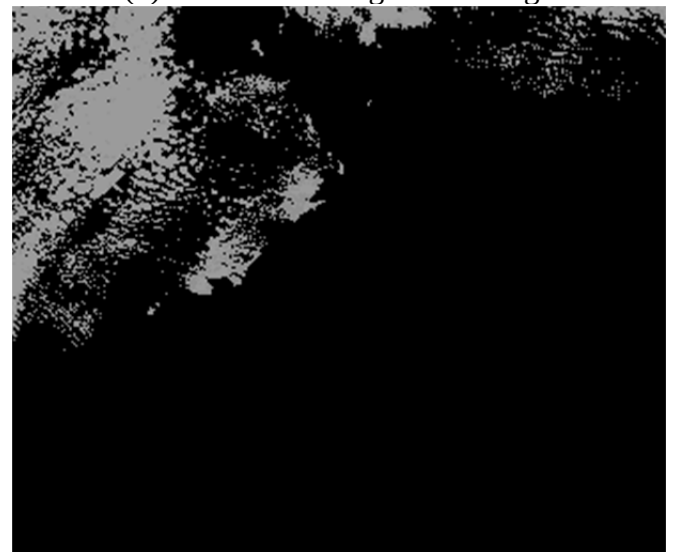
(a) VIIRS M10 Nighttime Image



(b) VIIRS DNB Nighttime Image



(c) VIIRS IICMO Cloud Mask



(d) NGDC Cloud Mask

Figure 4. Comparison of nighttime imagery over Basra area in Iraq collected on January 8, 2013 between 22:35:56 and 22:31:26 UTC. The VIIRS M10 is a shortwave infrared band and is very sensitive to high-temperature targets, while VIIRS DNB observes visible lights. By comparing (a) with (b), it is clear that bright circles in (b) are likely to be gas flares. Comparing cloud detection results, c) VIIRS IICMO cloud mask recognizes gas flares as cloud, while (d) NGDC cloud detection avoids such mis-identification.

The cloud masking algorithm developed for the DMSP OLS data compares brightness temperature in the OLS thermal band with a reference surface temperature. Values significantly colder than the reference surface temperature are considered clouds. With VIIRS data there are three longwave infrared (LWIR) bands: M14, M15, and M16. As depicted in Table 2, the M15

and M16 bandpasses straddle the first and second half of the OLS thermal band. The VIIRS M15 brightness temperature image was chosen to be the input into the cloud detection algorithm.

Table 2. VIIRS LWIR Bands as Compared to DMSP-OLS [5]

VIIRS Band	Spectral Range (um)	OLS Equivalent	Spectral Range (um)
M14	8.400 - 8.700	-	-
M15	10.263 - 11.263	OLS thermal band	10.300 - 12.900
M16	11.538 - 12.488		

The reference surface temperature is taken from the National Center for Environmental Prediction (NCEP) Global Forecast System (GFS) model runs. The GFS creates global surface temperature grids at 6 hour intervals, in resolutions of 0.5, 1.0, and 2.5 degree (Figure 5). Each 6 hour model run also has a series of forecasts at 3-hour intervals. For the NGDC cloud masking algorithm, the 0.5 degree resolution surface temperature grids for each of the 6 hour model runs, and the 3-hour forecasts are used. For each day, this yields 12 reference surface temperature samples at 0.5 degree resolution. More information on these datasets can be found on the NCEP GFS website <<http://www.nco.ncep.noaa.gov/pmb/products/gfs/>>.

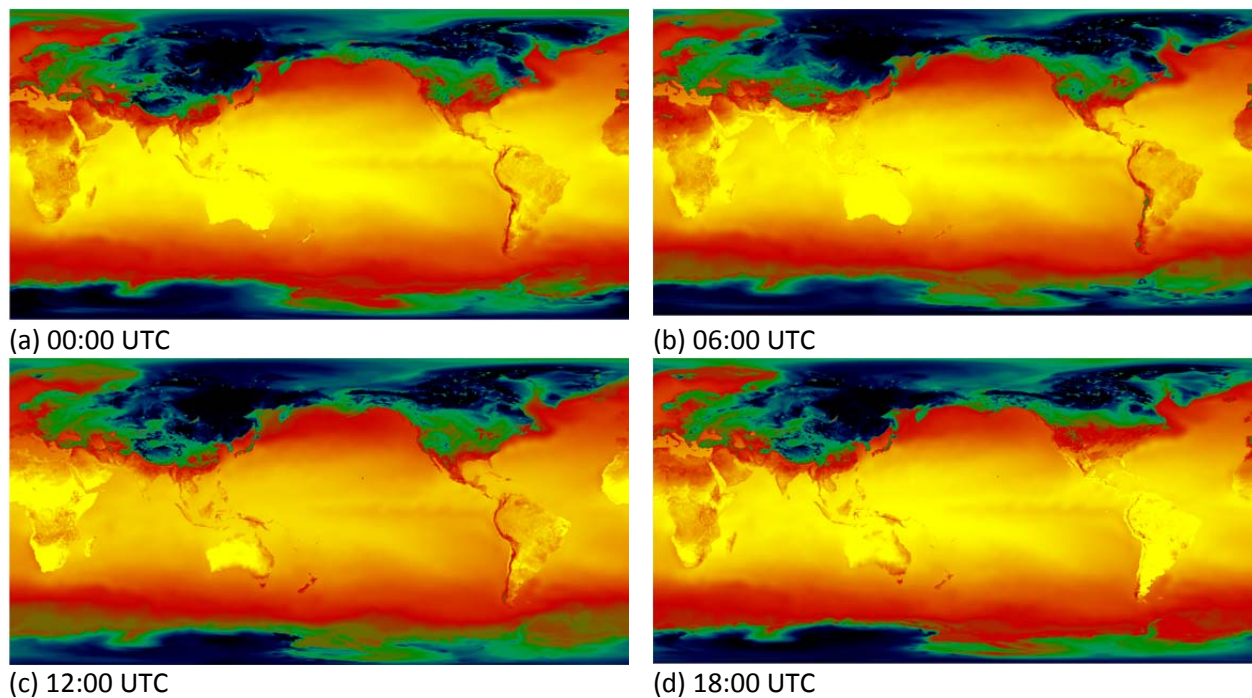


Figure 5. NCEP GFS 0.5 degree surface temperature grids for Jan. 18, 2012.

The NCEP GFS surface temperature grids are interpolated both spatially and temporally to match the VIIRS M15 data. At 0.5 degree resolution, the land/sea boundaries in the surface temperature grids are diffused as compared to the higher spatial resolution VIIRS M15 data

(742m at nadir). This results in the mis-identification of clouds along the land/sea interface. To address this issue, a land/sea lookup table was created for each 0.5 degree grid cell that covers a land/sea boundary. The lookup table identifies the location of the closest all-land and all-sea 0.5 degree grid cells. For each VIIRS M15 pixel a determination to use either the land or sea lookup table is made comparing the lat/lon position against a 30 arc-second land/sea mask. Then, based on the time of the VIIRS M15 scan, the temporally-adjacent GFS model runs and 3-hr predictions are identified and the land/sea lookup table is used to substitute temperature values if needed. Finally a reference temperature estimate is created by first bilinearly interpolating the land/sea adjusted surface temperature values to match the latitude and longitude of the VIIRS M15 pixels, and then by linearly time-interpolating the spatially adjusted surface temperatures as a function of the M15 scan time.

A thermal difference image is then calculated by subtracting the VIIRS M15 brightness temperature from the spatially and temporally-adjusted reference surface temperature estimate (Figure 6).

Clouds are identified by applying thresholds to temperature difference as shown in Table 3 and Figure 7.

Table 3 Cloud Identification Threshold

Mask Value	Threshold (deg K)
Cloud	< -13
Possible Cloud	-13 < and < -10
Clear	> -10

- Find surface temp. files before/after target aggregate
- Land/Sea Expansion by LUT

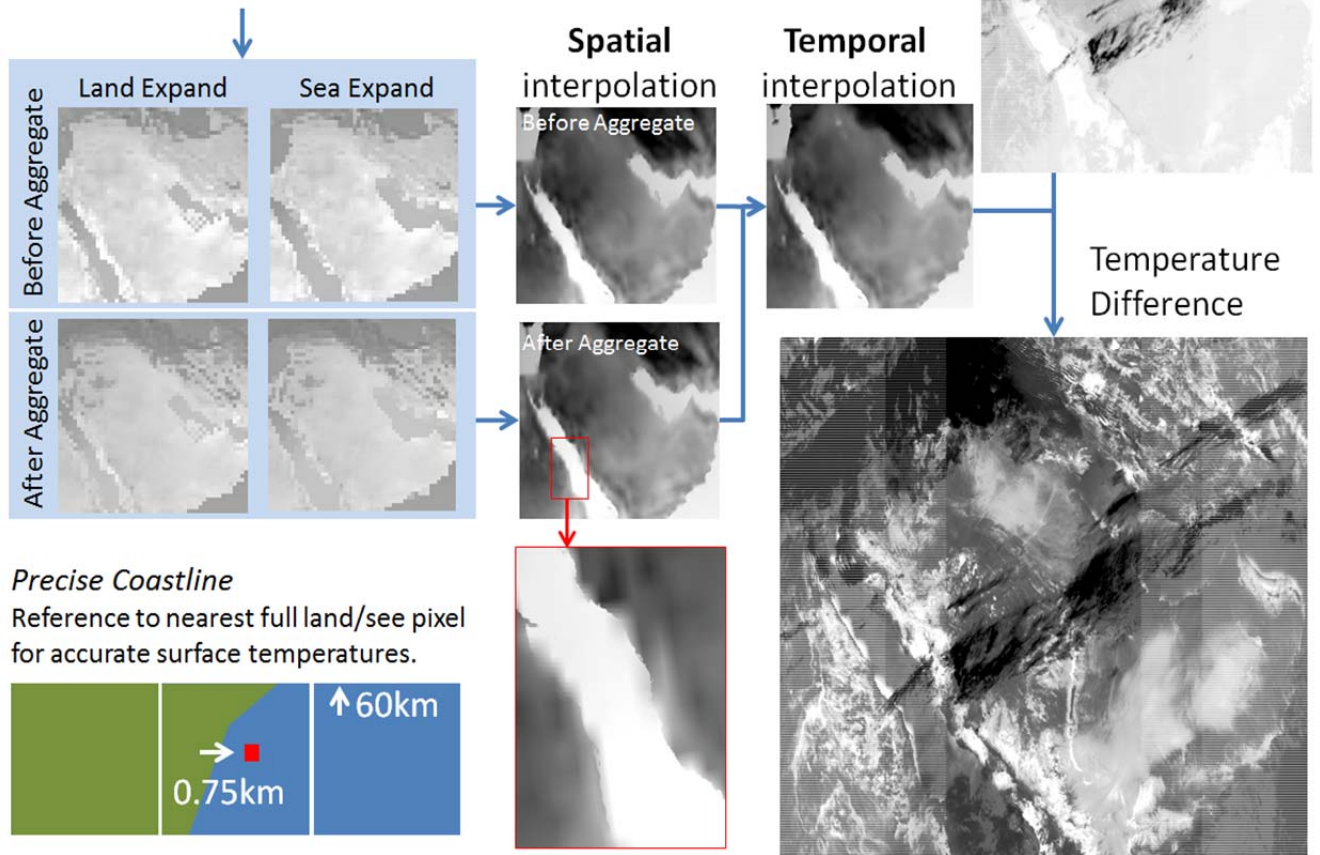
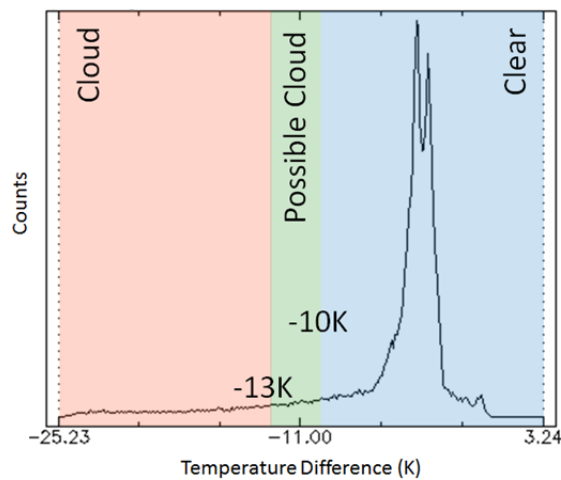
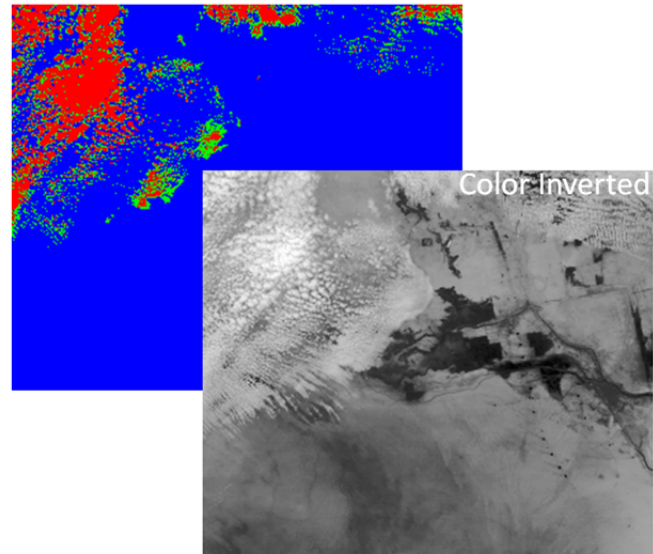


Figure 6. A graphical representation showing the algorithm for using a land/sea lookup table and spatial/temporal interpolation to compare NCEP GFS modeled surface temperature with the VIIRS M15 brightness temperature.



(a) Temperature Difference Histogram with Cloud Detection Thresholds shown.



(b) Cloud mask result compared with temperature difference image (color inverted to show cold areas in white). Red depicts Cloud flag set, green for Possible Clouds, and blue for no clouds present.

Figure 7. Threshold Application in NGDC's M15 Cloud Masking algorithm.

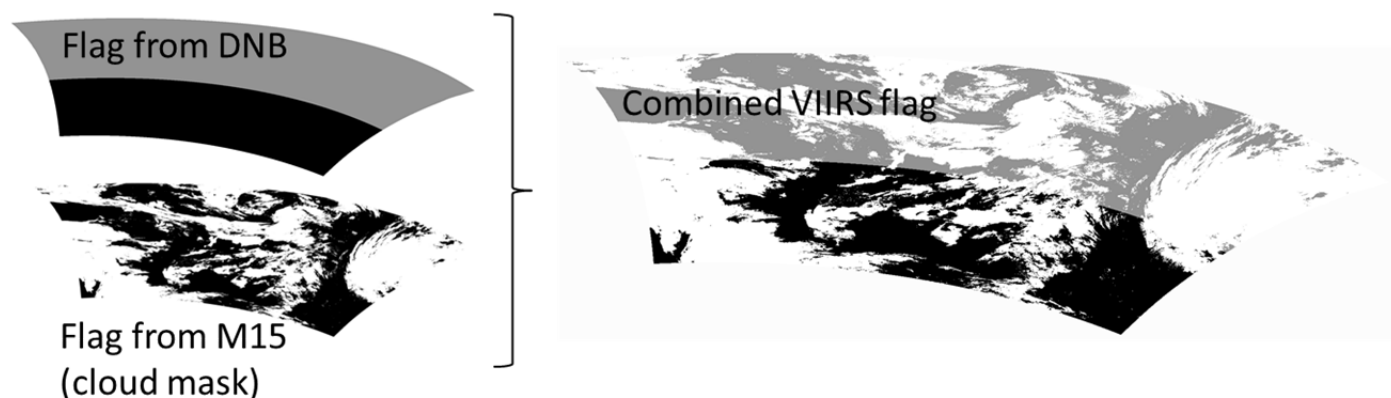
2.2. Reprojection

As just described, the flag image categories use information from both the VIIRS DNB and M15. The DNB and M-band detector arrays are slightly offset from each other on the VIIRS instrument and use different pixel aggregation schemes, therefore the flag images from the DNB and M15 are kept separate and combined into one flag image after reprojection.

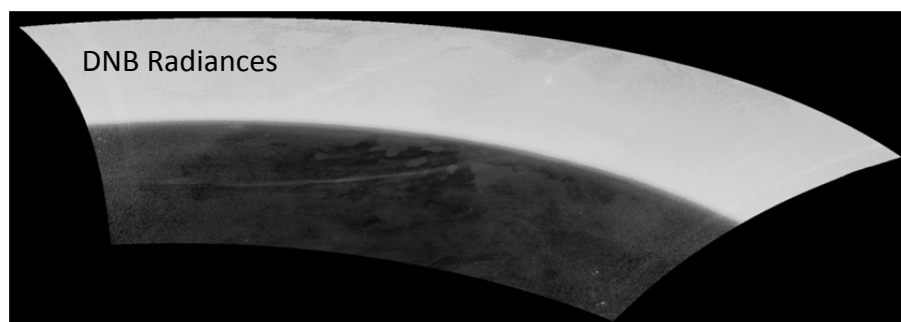
The reprojection software was created at NGDC specifically to work with the VIIRS data [6]. The VIIRS M-band data comes with an accompanying terrain-corrected geolocation file, so these lat/lon pairs can be used directly as input into the reprojection software. The DNB, however, is accompanied only with ellipsoid geolocation. So, NGDC created a terrain-correction module to work with the DNB data. The terrain-correction algorithm assigns new latitudes and longitudes to each DNB pixel based on ellipsoid geolocation, scan angle, satellite position, and a digital elevation model of the earth's surface. The reprojection software then uses the terrain-corrected geolocation values to resample the input data into a geographic grid using the nearest neighbor resampling technique.

Three different image types are reprojected into 15 arc-second grids, the DNB radiance images, the DNB flag images, and the M15 flag images. After reprojection, the DNB and

M15 flag images are combined into one “VIIRS flag” or vflag image (Figure 8). The reprojected DNB radiance and vflag images are input to the compositing algorithm.



a) Reprojection and merging of DNB and M15 flags to create one VIIRS flag image.



b) Reprojected VIIRS DNB radiance image

Figure 8. Reprojected VIIRS flag and DNB for use in compositing.

2.3. Making the VIIRS Nighttime DNB Composite

To create this preliminary composite, DNB radiances are included only if the following companion flag data bits are set as:

Daytime:	Off
Terminator Zone (twilight):	Off
Stray Light:	Off
Zero Lunar Illuminance:	On
Clouds:	Clear
No Data:	Off

The compositing process takes all input DNB radiance and vflag grids, masks each DNB radiance image using the flag image configured to values shown above, and creates a suite

of output files. This includes an average DNB radiance image, an image showing the number of cloud-free observations used, and a standard deviation DNB radiance image.

3. Results

Using the methods described here, the authors have processed two separate time periods and made VIIRS DNB composite products. The time ranges are April 18-26, 2012 and October 11-23, 2012. Shown in Figure 9 are the average DNB radiance product and the number of cloud-free observations for the October time period. Full-resolution crops of the average DNB composite for select geographic areas are shown in Figure 10. The composites for each of the time periods, along with a combined composite are available for download at http://www.ngdc.noaa.gov/eog/viirs/download_viirs_ntl.html.

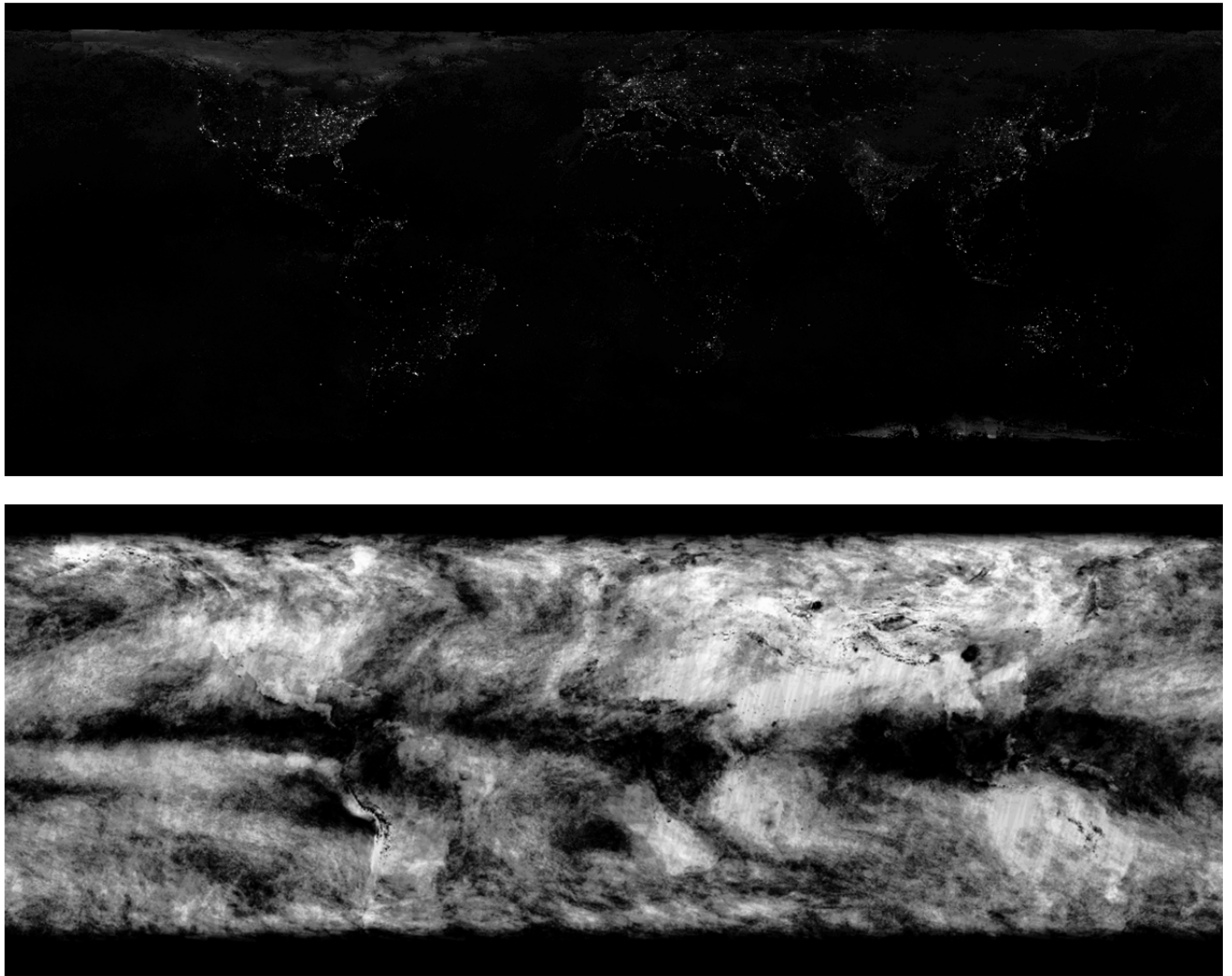


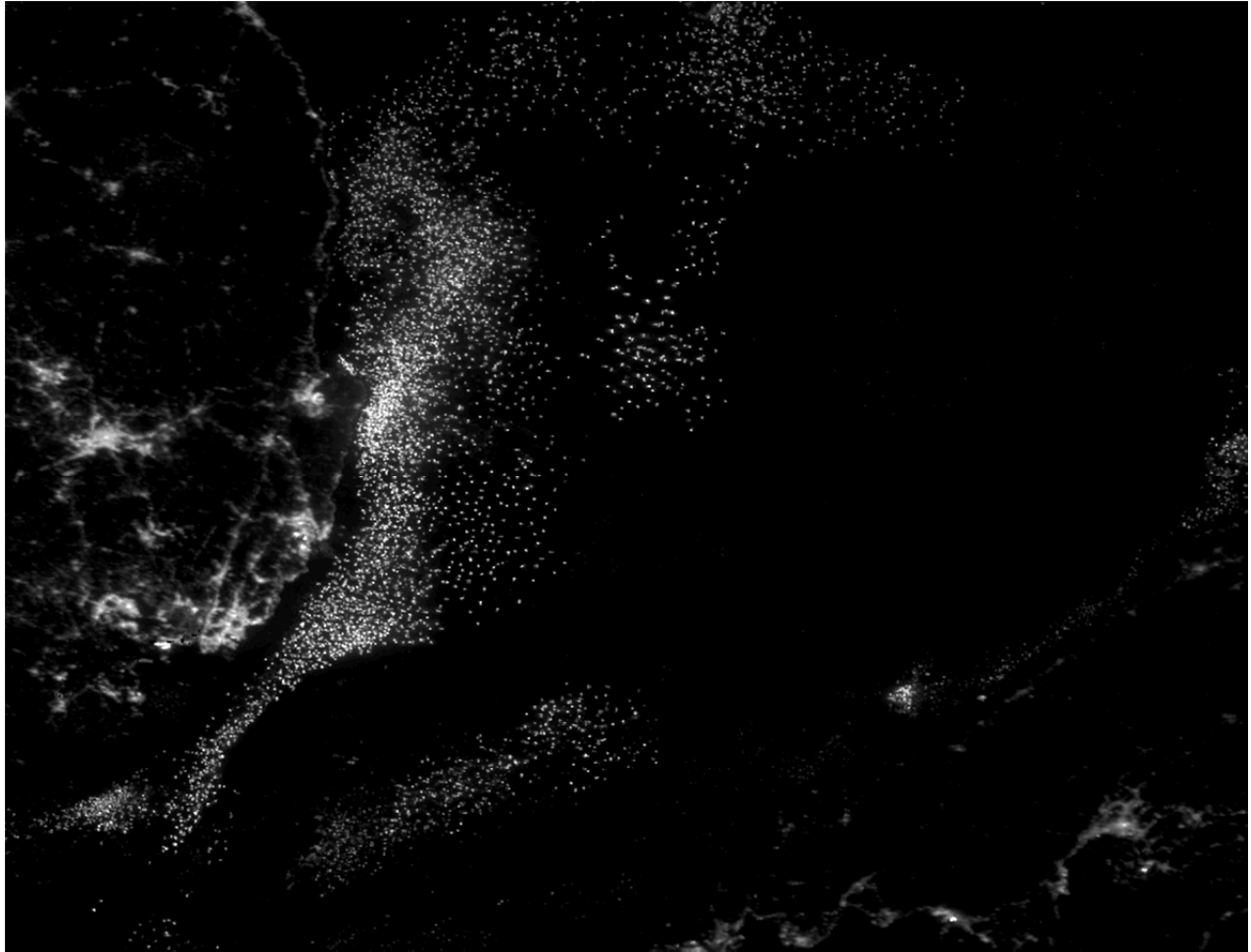
Figure 9: Composite products for October 11-23, 2012. a) Average DNB radiances (nanoWatts/cm²/sr). Bright areas over Canada and below Australia are due to lighting from aurora. b) Number of available high-quality cloud-free observations for October 11-23, 2012.



a) United Arab Emirates. The bright lights over the water are from gas flares.



b) Nile Delta



c) Lights from fishing boats in the Korea Strait and Sea of Japan.

Figure 10: Average DNB radiances for October 11-23, 2012 over select geographic regions.

4. Conclusions

The VIIRS DNB composite products will be a significant improvement over products the authors were able to make with the DMSP OLS data. The increase in spatial resolution and dynamic range of the VIIRS DNB as compared to the DMSP OLS [7] will allow the authors to expand their work to look at lighting patterns within urban centers. The VIIRS DNB is also a calibrated sensor, so mapping temporal change of lighting will be much easier and more reliable than with the DMSP OLS.

The authors are actively working on algorithms to improve the VIIRS DNB composite products. Current work includes geolocation improvement, or orbit to orbit alignment, to keep lighting features sharp in the composite. The authors are also working on algorithms to filter out lights from aurora, stray light, and lighting hits due to the South Atlantic Anomaly. Planned work includes improving the cloud-detection algorithm, and to filter out ephemeral lights from boats and fires to create a VIIRS DNB stable- or persistent-lights product.

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